Homework 05

INT3405

October 2, 2024

Collaboration: List the names of all people you have collaborated with and for which question(s). You may discuss problems with other students, but you must not view or copy another student's written answers. Each student is responsible for independently understanding, writing, and submitting their own work.

Acknowledgments: If you reference solutions from published sources, the web, or other textbooks, you must acknowledge these sources. It's acceptable to use existing proofs for guidance, but this must be noted, and you should first attempt to solve the problem on your own. All students are expected to fully comprehend the steps they include in their written work.

Exercise 1. [Reject option in classifiers] In many classification problems one has the option either of assigning x to class j or, if you are too uncertain, of choosing the **reject option**. If the cost for rejects is less than the cost of falsely classifying the object, it may be the optimal action. Let α_i mean you choose action i, for $i = 1 : C + 1$, where C is the number of classes and $C+1$ is the reject action. Let $Y = j$ be the true (but unknown) **state of nature**. Define the loss function as follows:

$$\lambda(\alpha_i|Y = j) = \begin{cases} 0 & \text{if } i = j \text{ and } i = j \in \{1, .., C\} \\ \lambda_r & \text{if } i = C + 1 \\ \lambda_s & \text{otherwise} \end{cases} \tag{1}$$

In other words, you incur 0 loss if you correctly classify, you incur λ_r loss (cost) if you choose the reject option, and you incur λ_s loss (cost) if you make a substitution error (misclassification).

1. Show that the minimum risk is obtained if we decide $Y = j$ if $p(Y = j|x) \geq p(Y = k|x)$ for all k (i.e., j is the most probable class) and if $p(Y = j|x) \geq 1 - \frac{\lambda_r}{\lambda_s}$; otherwise we decide to reject.

2. Describe qualitatively what happens as λ_r/λ_s is increased from 0 to 1 (i.e., the relative cost of rejection increases).

Exercise 2. Suppose we have a sample of N pairs x_i, y_i drawn i.i.d. from the distribution characterized as follows:

$$\begin{aligned} x_i &\sim h(x), \text{the design density} \\ y_i = f(x_i) + \epsilon_i, &\ f \text{ is the regression function} \\ \epsilon_i &\sim (0, \sigma^2) \text{ (mean zero, variance}\sigma^2) \end{aligned} \tag{2}$$

We construct an estimator for f *linear* in the y_i,

$$\hat{f}(x_0) = \sum_{i=1}^{N} \ell_i(x_0; \mathcal{X}) y_i, \tag{3}$$

where the weights $\ell_i(x_0; \mathcal{X})$ do not depend on the y_i, but do depend on the entire training sequence of x_i, denoted here by $\mathcal{X}$.

1. Show that linear regression and k-nearest-neighbor regression are members of this class of estimators. Describe explicitly the weights $\ell_i(x_0; \mathcal{X})$ in each of these cases.

2. Decompose the conditional mean-squared error

   $$\mathbf{E}_{\mathcal{Y}/\mathcal{X}}(f(x_0 - \hat{f}(x_0))^2$$

   into a conditional squared bias and a conditional variance component. Like $\mathcal{X}, \mathcal{Y}$ represents the entire training sequence of y_i.

Exercise 3. Write a pseudo of **Random Forest** regression algorithm.